Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY TECHNICAL BOOTCAMP · 40 HOURS

Data Science & Machine Learning

From data to deployed models — the complete ML engineering pipeline

5	40	20+	5
DAYS	HOURS	HANDS-ON LABS	ML PROJECTS

Course Overview

This 5-day hands-on bootcamp covers the complete data science and machine learning pipeline — from Python fundamentals and data wrangling through supervised and unsupervised learning to model evaluation, deployment, and real-world project delivery. Participants build 5 end-to-end ML projects using industry datasets.

Built for practitioners, not theorists. Every concept is immediately applied through hands-on coding labs. You'll work with real-world datasets from BFSI, healthcare, and e-commerce domains. By Day 5, you'll have a portfolio of 5 deployed ML models and the skills to tackle any data science challenge in your organization.

Who Is This For?

👨‍💻 **Developers**
Software engineers transitioning into ML and data science

📊 **Data Analysts**
Analysts ready to move beyond BI into predictive modeling

🎓 **Fresh Graduates**
Engineering and science graduates entering the AI field

⚙️ **IT Professionals**
IT teams building internal ML capabilities

Learning Outcomes

1. **Master Python for data science** — NumPy, Pandas, Matplotlib, Seaborn for data manipulation, analysis, and visualization
2. **Build supervised ML models** — linear regression, logistic regression, decision trees, random forests, gradient boosting (XGBoost)
3. **Apply unsupervised learning** — clustering (K-Means, DBSCAN), dimensionality reduction (PCA), and anomaly detection
4. **Engineer features like a pro** — data cleaning, imputation, encoding, scaling, feature selection, and pipeline construction
5. **Evaluate and tune models** — cross-validation, hyperparameter tuning, bias-variance tradeoff, confusion matrix, ROC-AUC
6. **Deploy ML models** — serve models via FastAPI, containerize with Docker, and deploy to Azure ML or AWS SageMaker

Prerequisites

Basic programming experience (any language). Python knowledge is helpful but not mandatory — Day 1 covers Python fundamentals. Laptop with Python 3.10+ installed.

5-Day Curriculum

Interactive format with live coding, hands-on labs, and project-based learning. Schedule and session order are fully customizable based on your organization's needs and time zones.

01 Python for Data Science & Exploratory Data Analysis

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A

Python Foundations for ML

- Python essentials: data types, control flow, functions, OOP
- NumPy: arrays, broadcasting, vectorized operations
- Pandas: DataFrames, indexing, groupby, merge, pivot tables
- Lab: Load, clean, and explore a real-world banking dataset
- Jupyter Notebooks and development environment setup

SESSION B

Data Visualization & EDA

- Matplotlib and Seaborn for statistical visualization
- Distribution plots, correlation heatmaps, pair plots
- Exploratory Data Analysis (EDA) methodology
- Lab: Complete EDA on a customer churn dataset
- Lab: Build an interactive dashboard with Plotly

02 Data Preprocessing & Feature Engineering

LIVE CODING + HANDS-ON LABS + WORKSHOP

SESSION A

Data Cleaning & Preprocessing

- Handling missing values: imputation strategies
- Outlier detection and treatment (IQR, Z-score)
- Encoding categorical variables: one-hot, label, target encoding
- Feature scaling: standardization, normalization, robust scaling
- Lab: Clean and preprocess a messy healthcare dataset

SESSION B

Feature Engineering & Pipelines

- Feature creation: domain-driven and automated approaches
- Feature selection: correlation, mutual information, RFE
- Scikit-learn Pipelines: chaining preprocessing + model training
- Lab: Build an end-to-end preprocessing pipeline
- Lab: Feature engineering for a credit scoring dataset

03 Supervised Learning — Regression & Classification

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A

Regression Models

- Linear Regression: assumptions, implementation, interpretation
- Polynomial Regression and regularization (Ridge, Lasso, ElasticNet)
- Model evaluation: MSE, RMSE, MAE, R^2 score
- Lab: Predict house prices using multiple regression techniques
- Lab: Sales forecasting model for an e-commerce dataset

SESSION B

Classification Models

- Logistic Regression: binary and multi-class classification
- Decision Trees and Random Forests
- Gradient Boosting: XGBoost, LightGBM fundamentals
- Evaluation: confusion matrix, precision, recall, F1, ROC-AUC
- Lab: Build a customer churn prediction model

04 Unsupervised Learning & Model Optimization

LIVE CODING + HANDS-ON LABS + WORKSHOP

SESSION A

Clustering & Dimensionality Reduction

- K-Means Clustering: elbow method, silhouette score
- DBSCAN and hierarchical clustering
- PCA for dimensionality reduction and visualization
- Lab: Customer segmentation using K-Means on retail data
- Lab: Anomaly detection in financial transactions

SESSION B

Model Tuning & Validation

- Cross-validation: K-Fold, Stratified, Leave-One-Out
- Hyperparameter tuning: GridSearchCV, RandomizedSearchCV, Optuna
- Bias-variance tradeoff and learning curves
- Ensemble methods: bagging, boosting, stacking
- Lab: Optimize a fraud detection model for production accuracy

05 NLP Fundamentals, Deployment & Capstone

CAPSTONE PROJECT + DEPLOYMENT LAB + PRESENTATIONS

SESSION A

NLP & Text Analytics Basics

- Text preprocessing: tokenization, stemming, lemmatization
- TF-IDF vectorization and word embeddings overview
- Sentiment analysis and text classification with Scikit-learn
- Lab: Build a product review sentiment classifier
- Introduction to Hugging Face Transformers for NLP

SESSION B

Model Deployment & Capstone

- Serving ML models with FastAPI
- Containerization with Docker
- Deployment to Azure ML or AWS SageMaker (demo)
- Capstone: End-to-end ML project — team presentations
- Certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools used hands-on with live coding and lab exercises

Python 3.x · NumPy · Pandas · Scikit-learn · XGBoost · LightGBM · Matplotlib · Seaborn · Plotly · Jupyter Notebooks · FastAPI · Docker · Hugging Face · Azure ML (demo) · AWS SageMaker (demo) · Optuna

What You Take Home

🤖 5 ML Projects
Complete Jupyter notebooks with code, data, and documentation — house price, churn, credit scoring, fraud detection, sentiment

📐 ML Pipeline Templates
Reusable Scikit-learn pipeline templates for preprocessing, training, evaluation, and deployment

📊 EDA Toolkit
Python scripts and notebook templates for rapid exploratory data analysis on any dataset

📋 Cheat Sheets
Quick reference cards for Pandas, Scikit-learn, feature engineering, and model evaluation metrics

Certification of Completion

Gennoor Tech — Certificate of Completion

Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in **"Data Science & Machine Learning"**. Digitally signed by the Founder & Director. Validates proficiency in Python data science, ML model building, feature engineering, model evaluation, and deployment.

Recommended Next Certifications

- **Microsoft AI-300**
  Machine Learning Operations (MLOps) Engineer Associate
- **Microsoft AI-103**
  Azure AI App & Agent Developer Associate
- **AWS MLS-C01**
  AWS Machine Learning Specialty
- **Google PMLE**
  Professional ML Engineer

Designed & Delivered By

Jalal Ahmed Khan

Founder & Director, Gennoor Tech Private Limited

Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of experience. Delivered 80+ ML and data science training programs for Microsoft, EY, IBM, Boeing, TCS, Capgemini, Wipro, and enterprise teams across GCC, Africa, and APAC regions.

16+ active Microsoft certifications including AI-300 (Azure Data Scientist), AI-103, AZ-400, PL-400, and GitHub certifications. Open-source contributor with ML pipelines on GitHub. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

🏢 **On-Site** — customized at your office | 💻 **Virtual Live** — interactive via Teams/Zoom | 🌍 **International** — GCC, Africa, APAC, UK

Ready to Build Intelligent Models?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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